TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN
KHTN
ĐHQG-HCM

Introduction to IT

2025

L2

Introduction to LaTeX

Learn LaTeX in 30 minute

What is LaTeX?

- LaTeX (pronounced "LAY-tek" or "LAH-tek") is a tool for typesetting professional-looking documents.
- Key difference:
  - Not WYSIWYG like Word
  - Works with plain text files containing LaTeX commands
  - TeX engine processes and produces professional PDF files
- Philosophy: You focus on content, LaTeX handles the presentation!

Why Learn LaTeX?

Advantages:

- Excellent support for complex mathematics and equations.
- Facilities for footnotes, cross-referencing, and bibliography management.

Why Learn LaTeX?

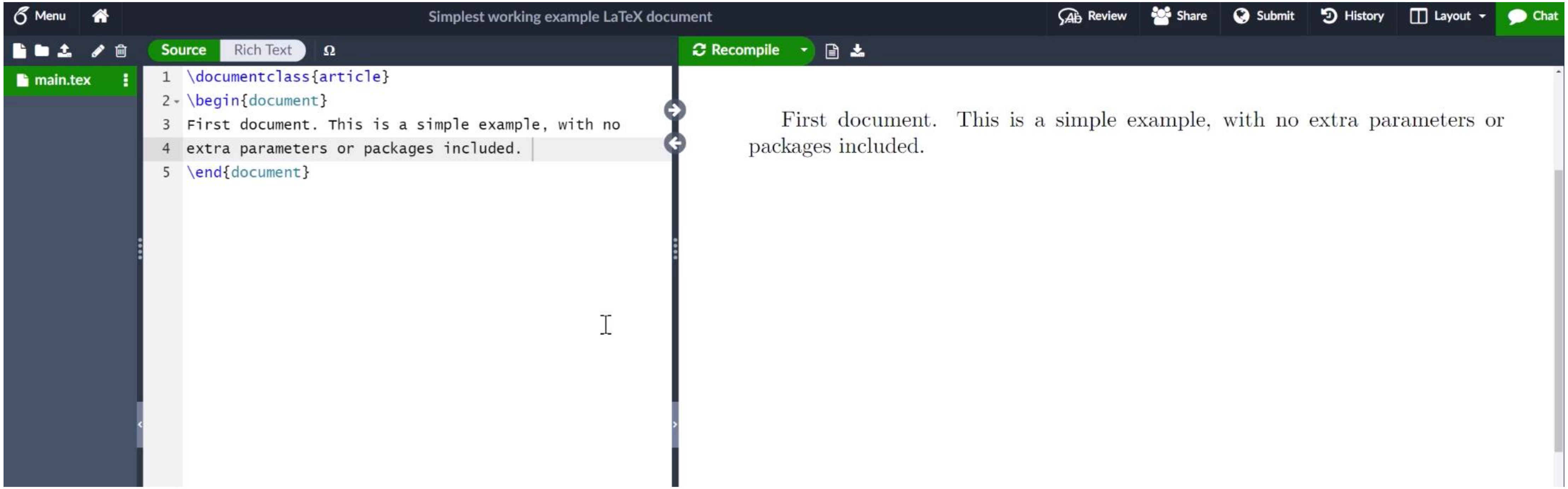
Advantages:

- Automated generation of table of contents, lists of figures, indexes.
- Highly customizable through thousands of free packages.
- Separation of content and style.

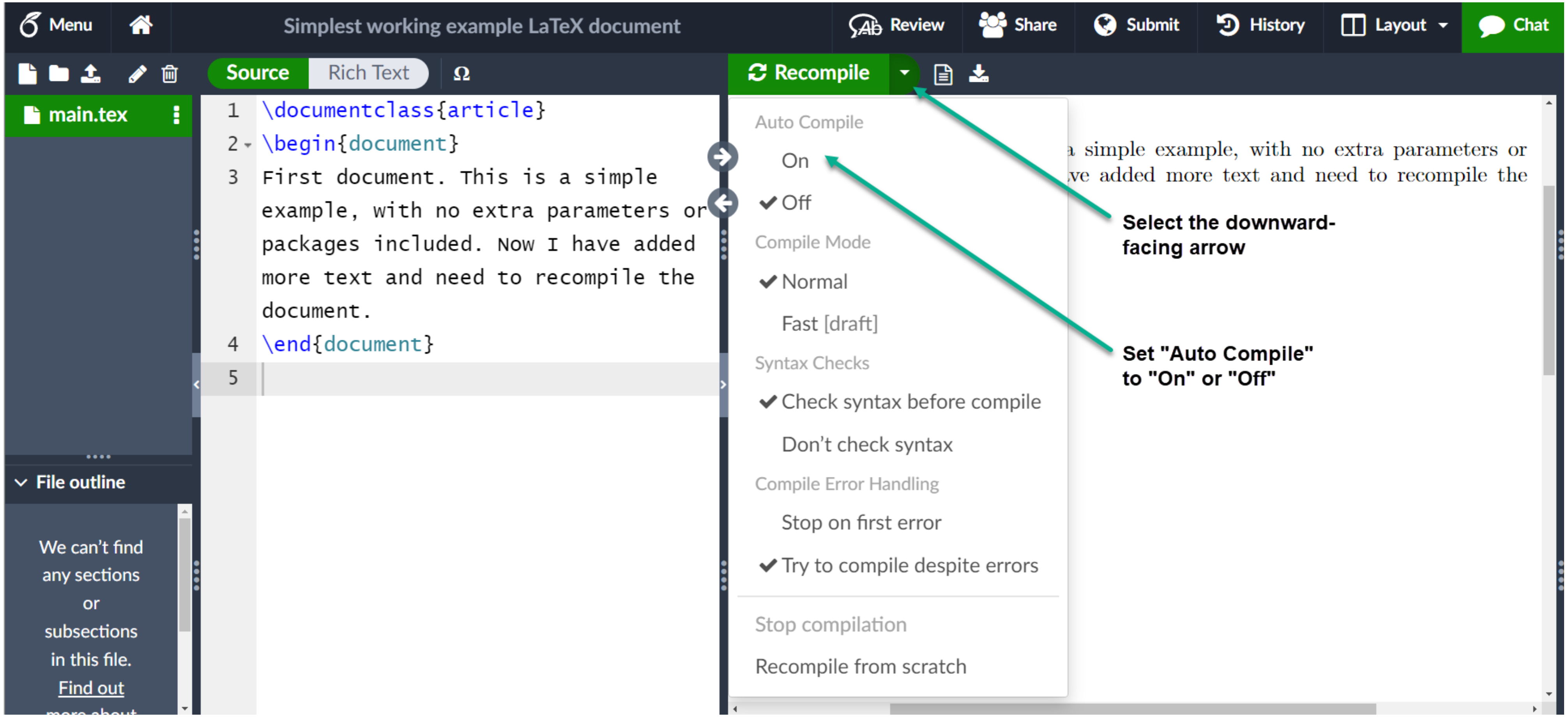
Your First LaTeX Document

```
\documentclass{article}

\begin{document}

First document. This is a simple
example, with no extra parameters or
packages included.

\end{document}
```

Your First LaTeX Document



Your First LaTeX Document



Document Structure: The Preamble

- **Preamble** is the "setup" section before `\begin{document}`.
- In the Preamble, you can: Configure document class; Load packages; Define title, author, date

Document Structure: The Preamble

```
\documentclass[12pt, letterpaper]{article}

\usepackage{graphicx}

\title{My first LaTeX document}

\author{Hubert Farnsworth\thanks{Funded by Overleaf.}}

\date{August 2022}
```

Creating a Title

```
\documentclass[12pt, letterpaper]{article}
\title{My first LaTeX document}
\author{Hubert Farnsworth}
\date{August 2022}
```

Creating a Title

```
\begin{document}

\maketitle

We have now added a title, author and
date!

\end{document}
```

Creating a Title

My first LaTeX document

Hubert Farnsworth*

August 2022

We have now added a title, author and date to our first LaTeX document!

Adding Comments

```
\begin{document}

\maketitle

We have now added a title, author and date!

% This line is a comment. It won't be typeset.

\end{document}
```

Text Formatting

```
Some of the \textbf{greatest}
discoveries in \underline{science}
were made by \textbf{\textit{accident}}.
```

Text Formatting

Basic Commands:

- `\textbf{...}`: **Bold text**
- `\textit{...}`: *Italic text*
- `\underline{...}`: <u>Underlined text</u>
- `\emph{...}`: Emphasis (context-aware)

Adding Images

```
\documentclass{article}
\usepackage{graphicx}
\graphicspath{{images/}}
```

Adding Images

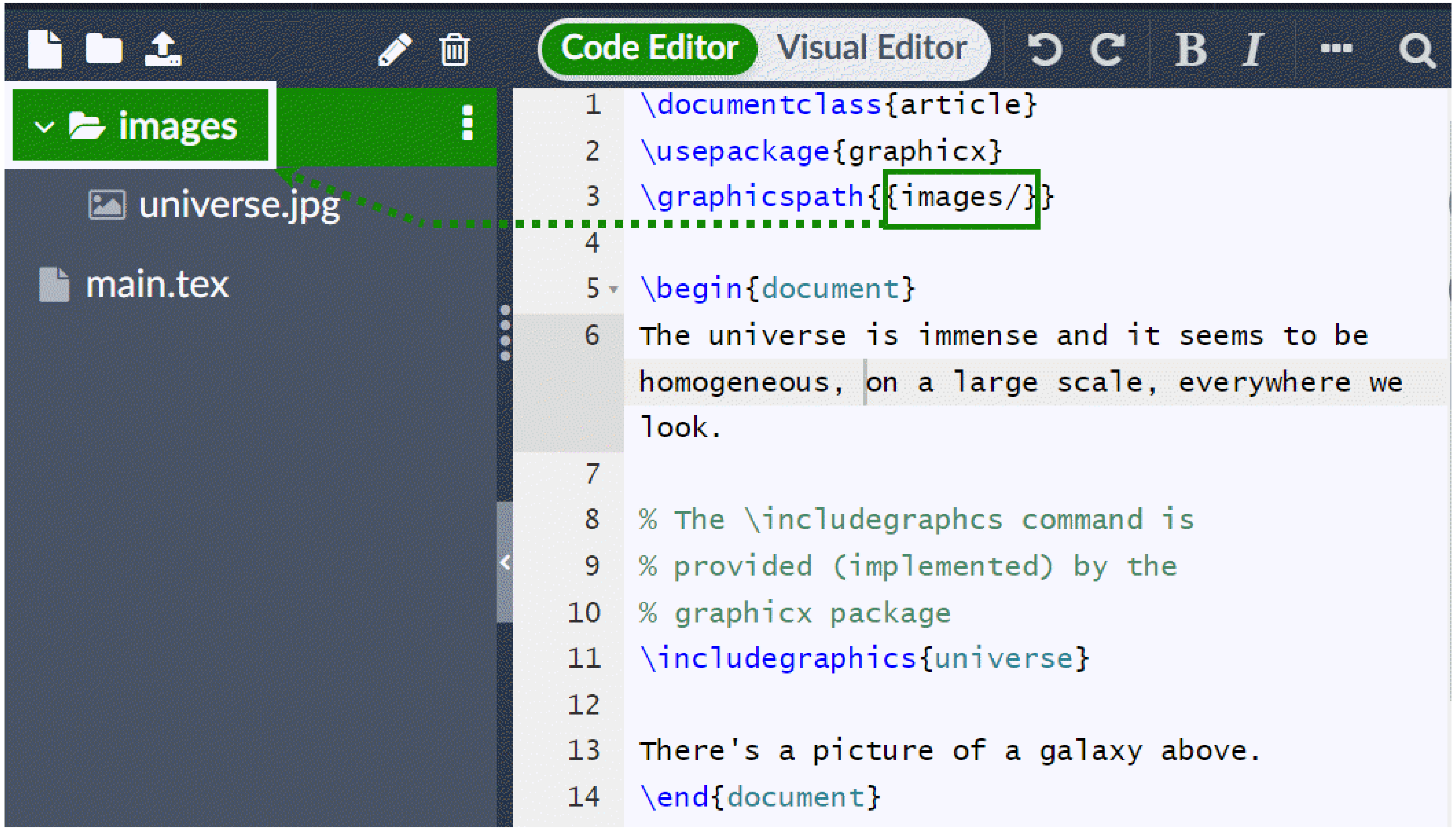
```
\begin{document}

The universe is immense...

\includegraphics{universe}

There's a picture of a galaxy above.

\end{document}
```

Adding Images

The universe is immense and it seems to be homogeneous, on a large scale, everywhere we look.

There's a picture of a galaxy above.

Adding Images



Adding Images

You need to use:

- `graphicx` package
- `\graphicspath{...}` command to specify directory
- `\includegraphics{filename}` to insert image

Captions and References

```
\begin{figure}[h]
    \centering
    \includegraphics[width=0.75\textwidth]
{mesh}
    \caption{A nice plot.}
    \label{fig:mesh1}
\end{figure}
```

Captions and References

```
As you can see in figure
\ref{fig:mesh1}...
```

```
This is on page \pageref{fig:mesh1}.
```

Captions and References

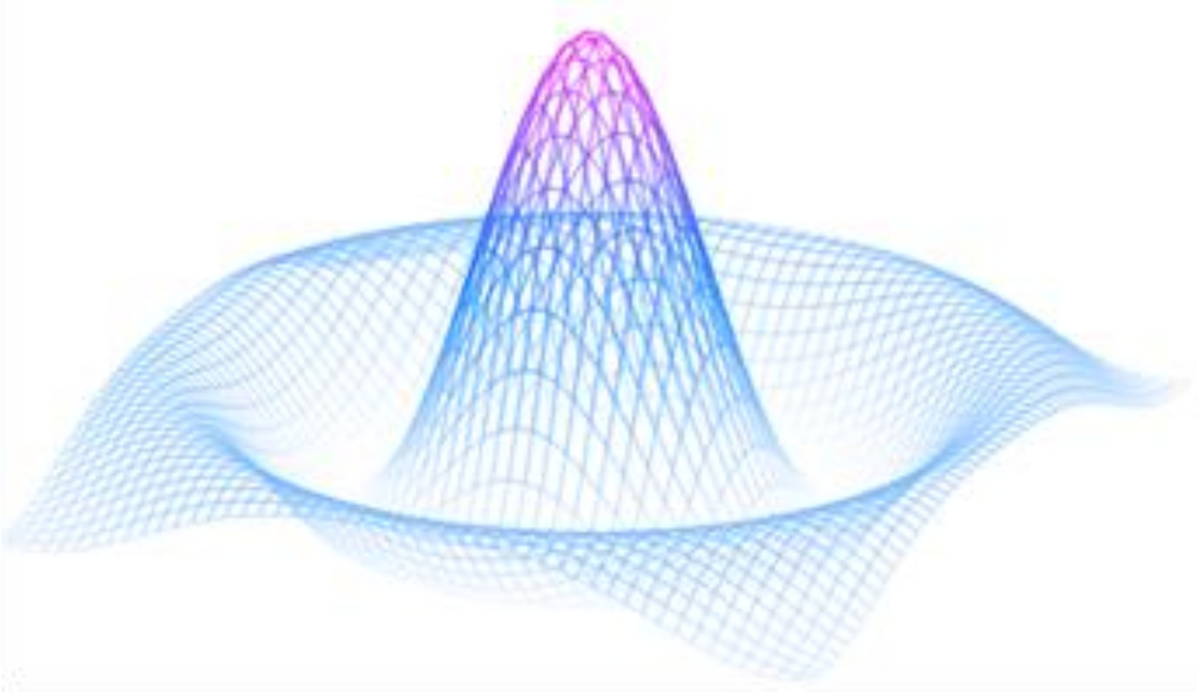


Figure 1: A nice plot.

As you can see in figure 1, the function grows near the origin. This example is on page 1.

Captions and References

Basic Commands

- `figure` environment: Wraps the image
- `\caption{...}`: Add caption
- `\label{...}`: Set label
- `\ref{...}`: Reference the figure

Unordered Lists

```
\begin{itemize}
    \item The individual entries are
indicated with a black dot.
    \item The text may be of any length.
\end{itemize}
```

Ordered Lists

```
\begin{enumerate}
    \item This is the first entry in our list.
    \item The list numbers increase with each entry.
\end{enumerate}
```

Mathematics: Inline Mode

In physics, the mass-energy equivalence is stated

by the equation $E=mc^2$, discovered in 1905.

Mathematics: Inline Mode

Ways to write inline formulas:

- `$ ... $`
- `\( ... \)`
- `\begin{math} ... \end{math}`

Mathematics: Display Mode

```
The mass-energy equivalence is described
by \[ E=mc^2 \]

discovered in 1905 by Albert Einstein.

\begin{equation}

E=m

\end{equation}
```

Mathematics: Display Mode

Display mode:

- `\[ ... \]`: Unnumbered equation

- `\begin{equation} ... \end{equation}`: Numbered equation

Advanced Mathematics Examples

```
Subscripts: $a_b$ and superscripts: $a^b$

\[ T^{i_1 i_2 \dots i_p}_{j_1 j_2 \dots j_q} \]

Integrals and fractions:

\[ \int_0^1 \frac{dx}{e^x} = \frac{e-1}{e} \]
```

Advanced Mathematics Examples

Greek letters: ω δ Ω Δ

Operators: $\sin(\beta)$, $\cos(\alpha)$, $\log(x)$

Document Structure

```latex
\documentclass{book}
\begin{document}
\chapter{First Chapter}
\section{Introduction}
\subsection{First Subsection}
\section*{Unnumbered Section}
\end{document}
```

Document Structure

Sectioning Basic Commands:

- `\part`, `\chapter` (book, report classes)
- `\section`, `\subsection`, `\subsubsection`
- Add `*` to remove numbering

Creating Basic Tables

```
\begin{center}
\begin{tabular}{c c c}
    cell1 & cell2 & cell3 \\
    cell4 & cell5 & cell6 \\
    cell7 & cell8 & cell9
\end{tabular}\end{center}
```

Tables with Borders

```
\begin{center} \begin{tabular}{|c|c|c|}
    \hline
    cell1 & cell2 & cell3 \\
    cell4 & cell5 & cell6 \\
    cell7 & cell8 & cell9 \\
    \hline\end{tabular}\end{center}
```

Tables with Captions

```
\begin{table}[h!]
\centering \begin{tabular}{||c c c||}
    \hline
    Col1 & Col2 & Col3 \\ [0.5ex]\hline\hline
    1 & 6 & 787 \\ \hline
\end{tabular}
```

Tables with Captions

```
\caption{Table to test captions.}
\label{table:data}
\end{table}
```

```
Table \ref{table:data} shows...
```

Automatic Table of Contents

```
\documentclass{article}
\title{Sections and Chapters}
\author{Your Name}
\date{\today}
```

Automatic Table of Contents

```
\begin{document}

\maketitle

\tableofcontents
```

Automatic Table of Contents

```
\section{Introduction}

This is the first section...

\section{Second Section}

More content here...

\end{document}
```

LaTeX Packages

- What is LaTeX Packages?

=> Packages extend LaTeX capabilities.

LaTeX Packages

- What is LaTeX Packages?

$\Rightarrow$Packages extend LaTeX capabilities.

```
\usepackage[options]{packagename}
```

LaTeX Packages

Example:

```
\usepackage{graphicx} % Images
\usepackage{amsmath} % Advanced math
\usepackage[utf8]{inputenc}% UTF-8
encoding
\usepackage{geometry} % Page layout
```

Abstracts

```
\documentclass{article}
\begin{document}
\begin{abstract}
This is a simple paragraph at the
beginning. A brief introduction about
the main subject.
\end{abstract}
```

Paragraphs and Line Breaks

- **Create new paragraph**: Press Enter twice (blank line)
- New line in same paragraph: `\\` or `\newline`

Paragraphs and Line Breaks

- **Create new paragraph**: Press Enter twice (blank line)
- New line in same paragraph: `\\` or `\newline`
- **Note:** Don't abuse `\\` for spacing!

Paragraphs and Line Breaks

```
First paragraph content here.


Second paragraph starts here.

Same paragraph but \\ new line here.
```

Tips and Best Practices

DO:

- Use appropriate document classes
- Load necessary packages in preamble
- Use `\label` and `\ref` for cross-references
- Comment your code for maintainability

Tips and Best Practices

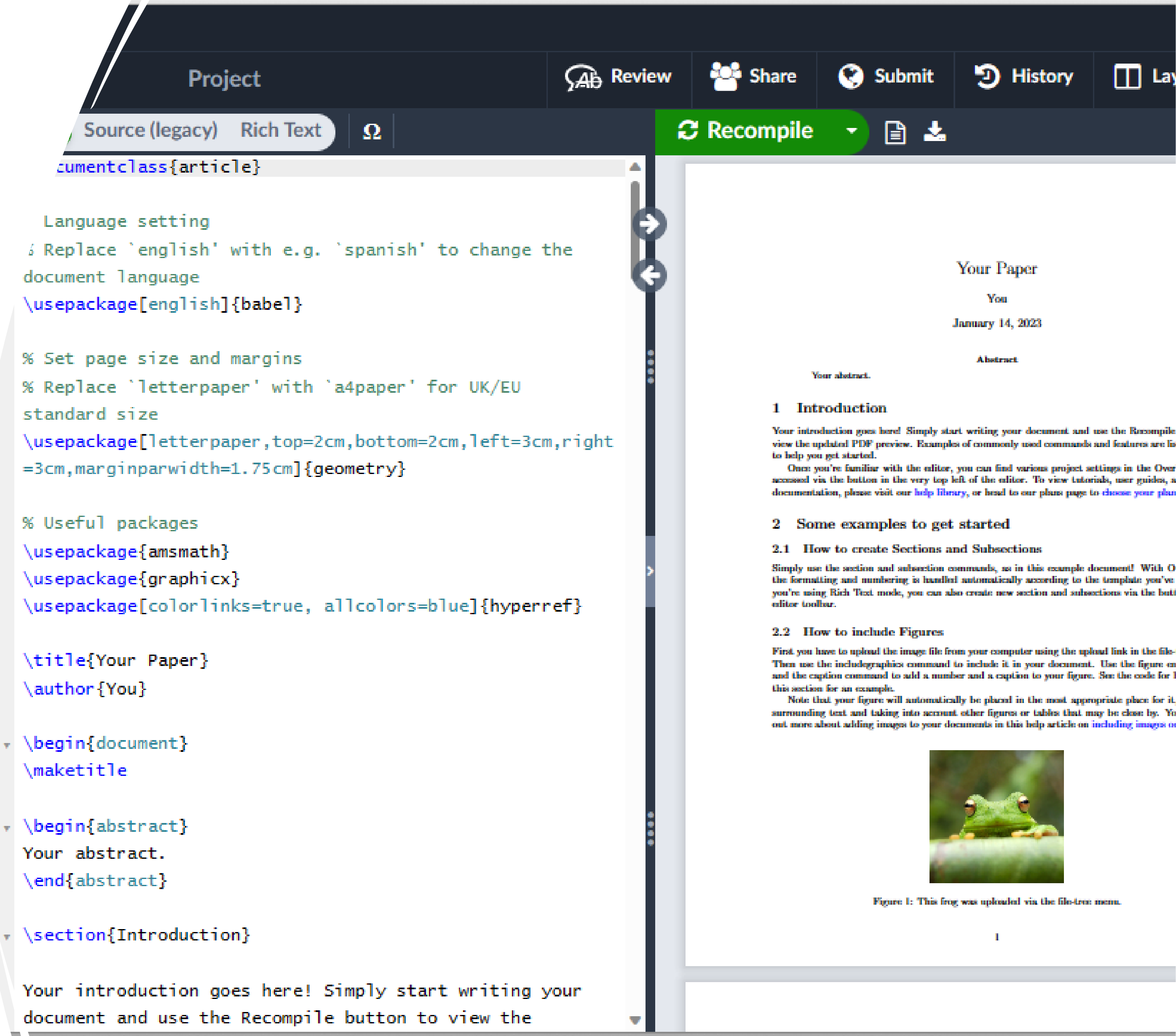
DONOT:

- Abuse `\\` for paragraph spacing
- Add extensions to image filenames in `\includegraphics`
- Use `$$` for display math (use `\[...\]` instead)

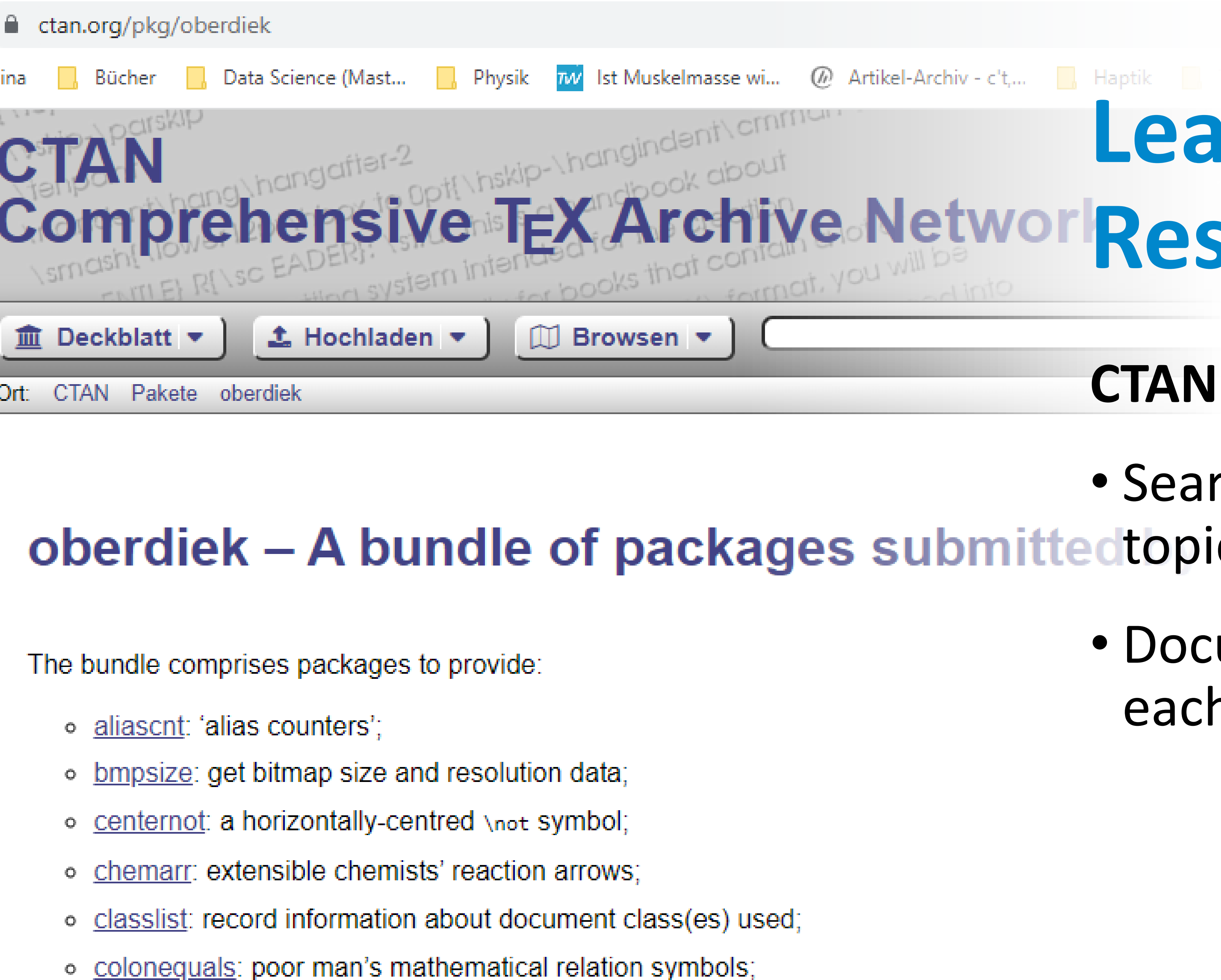
Learning Resources

Overleaf:

- Gallery with thousands of templates
- Detailed documentation
- Visual Editor and Code Editor



Learning Resources



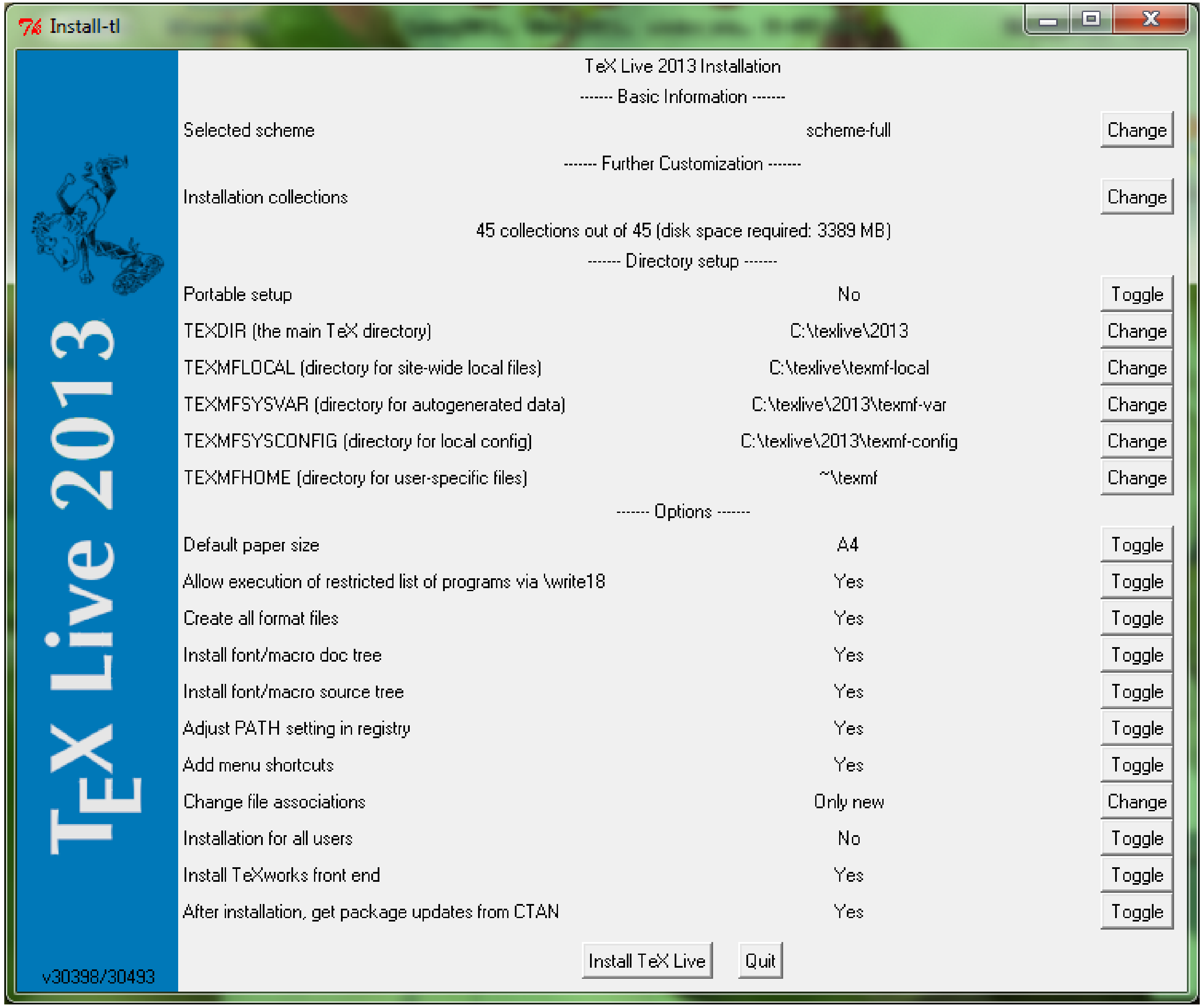
CTAN

- Search packages by topic
- Documentation for each package

Learning Resources

TeX Live:

- Complete LaTeX installation
- Annual updates



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